



Dell sued for \$4 million

Dell paid \$4 million to settle a dispute pending since 2008 against them for deceptive business practices



The magic of RGB LEDs

Very few displays use RGB LED technology but the advantage is a colour gamut that approaches 100% of the Adobe RGB colour space

HOW WE TESTED

We segregated notebooks into the following categories and subcategories:

Desktop Replacements

- Desktop replacements up to Rs. 40,000
- Desktop replacements between Rs. 40,001 and Rs. 60,000
- High-end desktop replacements (above Rs. 60,000)

Ultra Compacts

- Low power ultra compacts
- Ultra compacts

Corporate Notebooks

Apple Notebooks Gaming Notebooks

These categories need no explanations and four of them are simply based on usage patterns. We'd like to test Apple notebooks alongside other laptops but unfortunately most Windows based benchmarks we use have no OS X counterparts meaning an apples to apples (pardon the pun!) comparison just wasn't possible.

In terms of features we looked at and graded these notebooks

from every conceivable angle including the their hard drive size, type of optical drive, number of USB and FireWire ports, video connects, memory card reader, presence of E-SATA and even their weight. We also rated the build quality of each notebook and the ergonomics, layout and feel of the keypad. The usability and tracking accuracy of the trackpad was also rated. Presence of dedicated shortcut keys for common functions like multimedia controls, Wi-Fi on/off, Bluetooth and trackpad disable/enable was also scored.

In terms of performance we ensured a clean install of Windows Vista by uninstalling unnecessary software from these notebooks, but used the bundled OS itself. So if a notebook came with Windows XP preinstalled, we used that. System Restore and indexing were disabled before testing. All battery saving features were disabled and the notebook connected to the mains at all times except during the battery test (for obvious

reasons). A disk defragmenter was run on each notebook prior to testing and we used latest versions of ATI and NVIDIA display drivers.

PC Mark Vantage and 3D Mark Vantage are a little too taxing for many notebooks whose specifications are not up to the latest desktop PC hardware so we used earlier versions. PC Mark05 is our generic benchmark of choice because it has one of the most complete benchmark suites from testing the hard drive page file performance, to office related benchmarks, flash player performance, web page rendering, game and 3D testing and application startup. It has got to be one of the most comprehensive benchmarks around. 3D Mark 2006 was used to get an idea of the graphics and CPU subsystems performance. SiSoft Sandra 2009 Pro was used to get an idea of memory bandwidth. CineBench R10 gives a good indication of a CPUs rendering performance. DisplayMate

Video Edition was used to judge the quality and colour capabilities of each notebooks display. Finally Performance Test 7.0s inbuilt Wi-Fi test was used to judge the performance of the wireless component with theoretical data transfers. We hooked up a desktop PC using RJ45 connectivity to a wireless router and used this to connect to the each notebook for all wireless testing.

Our real world tests consisted of subjective tests for movie watching, web camera quality and inbuilt speaker quality. DivX Encode 7.0 was used to encode a 100 MB .VOB file to DivX and this was used to test the CPUs encoding capabilities. Sequential and assorted tests were done for judging the storage subsystems performance. For gaming performance we used Doom 3, one of the most scalable and hardware sensitive games around. For gaming notebooks we also used three other newer games – *Crysis*, *S.T.A.L.K.E.R. Clear Sky* and *Company Of Heroes: Tales Of Valor*.



Apple MacBook Pro 15-inch

recommend the Macbook Pro 13-inch priced at Rs. 74,400 – good value for money and a very well built, great looking piece of equipment. If you want something really powerful we recommend the Apple Macbook Pro 15.4-inch – class-leading hardware and sublime looks and built, though its pricey at Rs. 1,41,600.

BUSINESS NOTEBOOKS: Tie and Suit, ready to boot

We've often asked ourselves – what makes a business notebook? Obviously something

compact, since businessmen on the go like to travel light. Something that's rugged and not too flashy – businessmen want productivity and usability more than pizzazz. Obviously performance, since time is a luxury few corporate users have. Backup software and other office productivity related tools are a must as well since the key to effective productivity is prevention of delays. Therefore a good business notebook should have all these features.

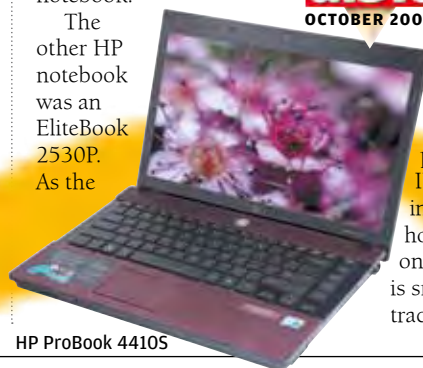
The four corporate-oriented products we received certainly seemed to meet all the parameters set. HPs ProBook 4410S was the first candidate up for test. It's got a nice textured finish on the lid and the dull maroon colour is really smart looking. Inside the notebook is a mix of glossy and matte black. It's compact and light but a little on the thick side. The keypad is segregated with a little less spacing that what Sony uses and owing to this the keypad is more usable – a solid plus for HP. The 4410S

comes with a complete backup and recovery suite from HP meaning your data is always safe. It's got a very mediocre display in terms of colour and while businessmen don't need colour accuracy and contrast it would be nice to have a slightly better screen. The trackpad works well and the Wi-Fi shortcut switch is usable. To be honest this ones price surprised us – we expected it to be well in excess of Rs. 50,000 but HP makes it available for as little as Rs. 40,000 – a solid buy for someone looking at a solidly built, compact, no frills notebook.

The other HP notebook was an EliteBook 2530P. As the

name suggest (i.e. *Elite* is higher than *Pro*) the 2530P is meant for the corporate with deeper pockets. It's even smaller with a 12.1-inch display. The lid has a brushed aluminium finish and this goes well with the grey-black body. The notebook is quite compact but feels sturdy. HP states the use of certain materials that increase reliability even in dusty conditions and this notebook is built to withstand wider variations in temperature.

Their DriveGuard technology that protects the hard drive in case of a sudden impact is also incorporated on this model. The processor is an Intel SU9300 – this series is known for its ultra low power consumption. An IBM-style track point is also included although to be honest hardly anyone uses one these days. The trackpad is small but works well and tracking is accurate and lag



HP ProBook 4410S